

AARFLingo: An Observational System and Evaluation Protocol for Multimodal Canine Intent Forecasting

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Preprint. Not a claim of 95% home-dog accuracy.

Abstract

Owners and researchers need a forecast of what a dog is about to do, not a shock after the fact. AARFLingo predicts a coupled triad—intent $\times$ emotion $\times$ surface behavior—from a room camera, an optional vocal encoder, and a notify-only neck-worn sensor puck. A 73-dimensional perception vector, stacked over 15 frames, feeds a multi-head classifier whose illegal combinations are rejected by an ethogram coupling matrix. The wearable is an ESP32-S3 puck (inertial measurement unit, I2S microphone, dual-wavelength neck photoplethysmography, skin thermistor) with no writable actuation characteristic. We report only numbers that exist in the companion repository. Stanford Dogs breed top-1 is 74.6% on public photographs. Barkopedia-held-out vocal accuracy is 34.5% on 298 real clips; synthetic vocal accuracy (80.9%) does not count. Triad and vitals checkpoints reach validation accuracy 1.0 on synthetic rows; those scores do not count toward v1.0. The v1.0 field bar is dog-held-out intent accuracy ≥ 0.95 on at least three home dogs. As of this writing the home eval file is empty, so the bar is unmet. The contribution of this preprint is the system, the ethics constraint, and the evaluation protocol a later labeled set must pass.

1 Introduction

Canine affect work often stops at a single axis (arousal, “happy/sad”) or a single sensor. Intent is a destination: the door, the toy, the bowl, withdrawal. AARFLingo treats intent, emotion, and behavior as one structured prediction so a play-bow cannot be paired with rest, and a door-oriented freeze can be read as “wants out” rather than as generic anxiety.

Three design choices follow. First, the label space is an ethogram: ten intents, seven emotions, and ten behaviors, with a coupling matrix that names allowed triples and forbidden pairs. Second, the wearable is observational. The Rev-A collar streams inertial, acoustic, and neck-photoplethysmographic features at 1 Hz over a notify-only GATT map. Firmware and schematic tests fail if shock, vibe, motor, or solenoid nets appear. Inference stays on a laptop, phone, or Jetson-class hub; the puck does not run a detector. Third, the accuracy bar is dog-held-out. Random-frame splits and synthetic validation accuracy can be perfect while every new dog fails. `scripts/vl_gate.py --require-bar` is the test a later home set must pass. It fails today.

Protocol files that a reviewer can run without this prose live in `docs/paper/` (`METHODS.md`, `generated RESULTS.md`, `DATASHEET.md`, `reproduce.md`). The repository is <https://github.com/Team-Deepiri/deepiri-aarflingo>.

2 Related work

Khosla et al. released Stanford Dogs (120 breeds, 20,580 photographs) as a fine-grained categorization set [Khosla et al., 2011]. We use it to pre-train a MobileNetV3 breed head (held-out top-1 74.6%) and as a source of real feature rows through the perception pipeline. Runtime detection is YOLOv8n, class dog, with an OpenCV motion fallback [Jocher et al., 2023].

Barkopedia and DogSpeak provide labeled canine vocalizations [Arlington Computational Linguistics, 2024a,b]. Our vocal encoder is trained on all 298 real Barkopedia clips available in-tree plus synthetic fillers; checkpoint selection uses real held-out accuracy (34.5%).

PhysioZoo publishes canine electrocardiograms with R-peaks and heart-rate variability annotations [Behar et al., 2018]. The in-repo vitals encoder is shaped after those corpora; the shipped checkpoint was trained on synthetic windows and reports validation accuracy 1.0, which we treat as a smoke test.

Quaranta et al. showed that tail laterality tracks approach versus withdrawal [Quaranta et al., 2007]. Ren et al. measured attractor-like wag dynamics [Ren et al., 2022]. DogFACS catalogs facial action units for dogs [Waller et al., 2013]. AARFLingo reserves named slots for those quantities; they are zeros when the camera cannot see the dog. Lee’s τ is the distance-over-closing-rate used as an approach feature toward annotated gaze zones [Lee, 1976].

We do not claim a prior field study of this triad on home dogs. That study is the v1.0 bar, and it has not been run.

3 Ethics and scope

AARFLingo is observational. It does not automate punishment, vibration, or restraint. The GATT map is notify-only. Owner opt-in is required. Clinic recordings need an institutional review board or equivalent. Delete-on-request is 30 days.

Collar keys named `arousal`, `still`, and `red` are autonomic or contact correlates. They are not valence, core temperature, cortisol, or SpO_2 . The red LED path is a second photoplethysmographic perfusion index, not a blood-gas measurement. Low-confidence or forbidden triples default to human review.

4 System

A live session is a loop. A USB or MIPI camera produces frames. Perception emits a 55-dimensional visual base vector (presence, box, gaze zones, approach geometry, optional tail, face, and gait). Optional `POST /infer/audio` writes vocal arousal, valence, and bark probability. Optional `POST /infer/collar` maps the puck’s 1 Hz CBOR map onto the existing electrocardiogram and inertial slots—no fourth wire protocol, and no new feature dimension. Cross-modal synchrony (phase-locking value, lag correlation, a two-state Kalman fusion) fills `sync_*` when both streams are present. Fifteen frames are flattened and scored by TriadNet. A coupling gate returns pass, review, or reject. A human can correct the triad; `POST /live/retrain` fine-tunes from that log.

The Jetson image is an L4T hub. It does not run on the ESP32.

5 Ethogram and coupling

Intents: approach, avoid, solicit-play, rest, guard-resource, explore, alert, outside, play, food. Emotions: calm, content, excited, anxious, fearful, frustrated, conflicted. Behaviors: loose tail-wag, tucked tail, play-bow, lip-lick, whale-eye, hard stare, yawning, sniff-ground, freeze, bark.

The coupling matrix lists weighted triples (play + excited + play-bow at 0.9) and forbidden pairs (rest + play-bow; avoid + play-bow; rest + excited). Training adds a coupling loss $\lambda \mathcal{L}_c$ with default $\lambda = 0.3$. At inference the gate rejects forbidden pairs and only passes a listed triple when mean-head confidence is at least 0.55.

6 Features

Let $x_t \in \mathbb{R}^{73}$ be the concatenated vector at frame t . The first 55 names are visual; the last 18 are modality outputs. A window of $T = 15$ frames is left-padded and flattened to $\mathbb{R}^{1095}$.

For each zone z with center c_z , dog position p and velocity v ,

$$d_z = \|c_z - p\|, \quad \hat{u}_z = (c_z - p)/d_z, \quad \dot{d}_z = v \cdot \hat{u}_z. \quad (1)$$

Closing rate is a clamped $\dot{d}_z$. Time-to-contact is $\tau_z = d_z / \max(\dot{d}_z, \varepsilon)$, inverted against a 60-frame horizon so that a large τ_z means imminent arrival. Heading is $\hat{v} \cdot \hat{u}_z$.

The 1 Hz collar map is projected onto slots the triad already has (`hr_bpm/180`, `rmssd_ms/150`, arousal, IMU RMS, still; all clamped to $[0, 1]$). `scripts/collar_listen.py` with `--runtime` posts each notify to `POST /infer/collar`.

7 Models

Shipped TriadNet is a two-layer multilayer perceptron (hidden 128, dropout 0.15) over the flattened window, with three linear heads. Confidence is the mean of the three selected softmax masses. Margin is the mean top-1 minus top-2 gap; a small margin is treated as review even when confidence is high.

`TriadNetTemporal` is a bidirectional LSTM with Bahdanau attention and extra arousal/valence heads. It is separately trainable and is not the default, because promoting it would break the shipped checkpoint layout.

If no checkpoint loads, gaze zones and motion pick a triple. That fallback is a bring-up aid, not a reported model.

8 Wearable

Rev-A board: 40×32 mm. Microcontroller ESP32-S3-MINI-1. Inertial measurement unit Bosch BMI270 on I2C (0x68). Microphone INMP441 on I2S. Neck photoplethysmography Texas Instruments AFE4404 (0x58) with infrared and 660 nm emitters. Contact temperature is a $10 \text{ k}\Omega$ $\beta 3950$ thermistor. Charge default is 100 mA. Firmware 0.3.0 advertises `aarf-collar`, requests MTU 247, and notifies a 23-key CBOR map at 1 Hz. The printed circuit is not yet signal-routed and is not fab-ready.

9 Evaluation protocol

The only number that can satisfy v1.0 is dog-held-out intent accuracy on a home set: at least three dogs; split by `dog_id`; labels from the ethogram; rows in `data/dog/eval/dog_split.jsonl`; scored by `python3 scripts/v1_gate.py --require-bar`. Target: accuracy ≥ 0.95 and macro-F1 reported beside it. Camera-only rows are valid; an optional `collar` object is counted but does not by itself meet the bar.

What does not count: synthetic triad or vitals `best_val_acc`, mixed vocal accuracy, Stanford Dogs breed top-1, and collar proxies used as intent labels.

Table 1: Honest scoreboard. Generated rows come from `docs/paper/RESULTS.md`.

Metric	Value	Counts toward v1.0?
Home dog-split accuracy	—	yes
Home dogs / rows	0 / 0	yes
Rev-A puck contract	ok	yes
Jetson hub image	ok	yes
Triad <code>best_val_acc</code> (synthetic, $n=40$)	1.0	no
Vocal real held-out (298 Barkopedia clips)	0.345	no
Vocal synthetic held-out	0.809	no
Vitals <code>best_val_acc</code> (synthetic)	1.0	no
Breed top-1 (Stanford Dogs)	0.746	no
<code>bar_met</code>	false	

10 Results

Table 1 is the generated report from `python3 scripts/v1_gate.py` on 21 August 2026, plus public-corpus context from in-repo metric files.

The blocker is the missing home JSONL. The vocal encoder beats chance on 298 real clips. The breed head is a public-photo number. The triad checkpoint memorizes its synthetic draw. None of those sentences is a home-dog result.

11 Limitations

The home set does not exist yet. Named visual features (facial units, Lyapunov wag, gait phase) are estimated from a 2D box and a lightweight pose. Collar heart-rate and RMSSD assume a still neck; motion and fur will corrupt peaks. Gyro RMS is not a behavior classifier. Package temperature is not skin; skin is not core. `TriadNetTemporal`, `PhysioZoo`-trained vitals, and a routed board are unfinished. The 95% target may be too high for dog-held-out intent with $N = 3$; the gate will say so if the labeled file exists and falls short.

12 Reproducibility

From a clone, without a GPU:

```
python3 scripts/v1_gate.py
python3 scripts/v1_gate.py --require-bar
./scripts/train_aarflingo.sh
python3 scripts/verify_artifacts.py
```

The last command checks that a checkpoint loads. It is not the v1.0 bar.

13 Conclusion

AARFLingo is an observational triad forecaster with a room camera, a vocal encoder, and a notify-only neck puck, scored by a dog-held-out gate that synthetic accuracy cannot pass. The software and the Rev-A

contract are in the repository. The home-dog measurement is not. That is the next experiment, not a number we will invent.

Acknowledgements

Public corpora: Stanford Dogs, Barkopedia, DogSpeak, AudioSet (whimper), PhysioZoo, Mendeley inertial sets. No animal was stimulated by this system. Code is Apache-2.0 [Deepiri, 2026].

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